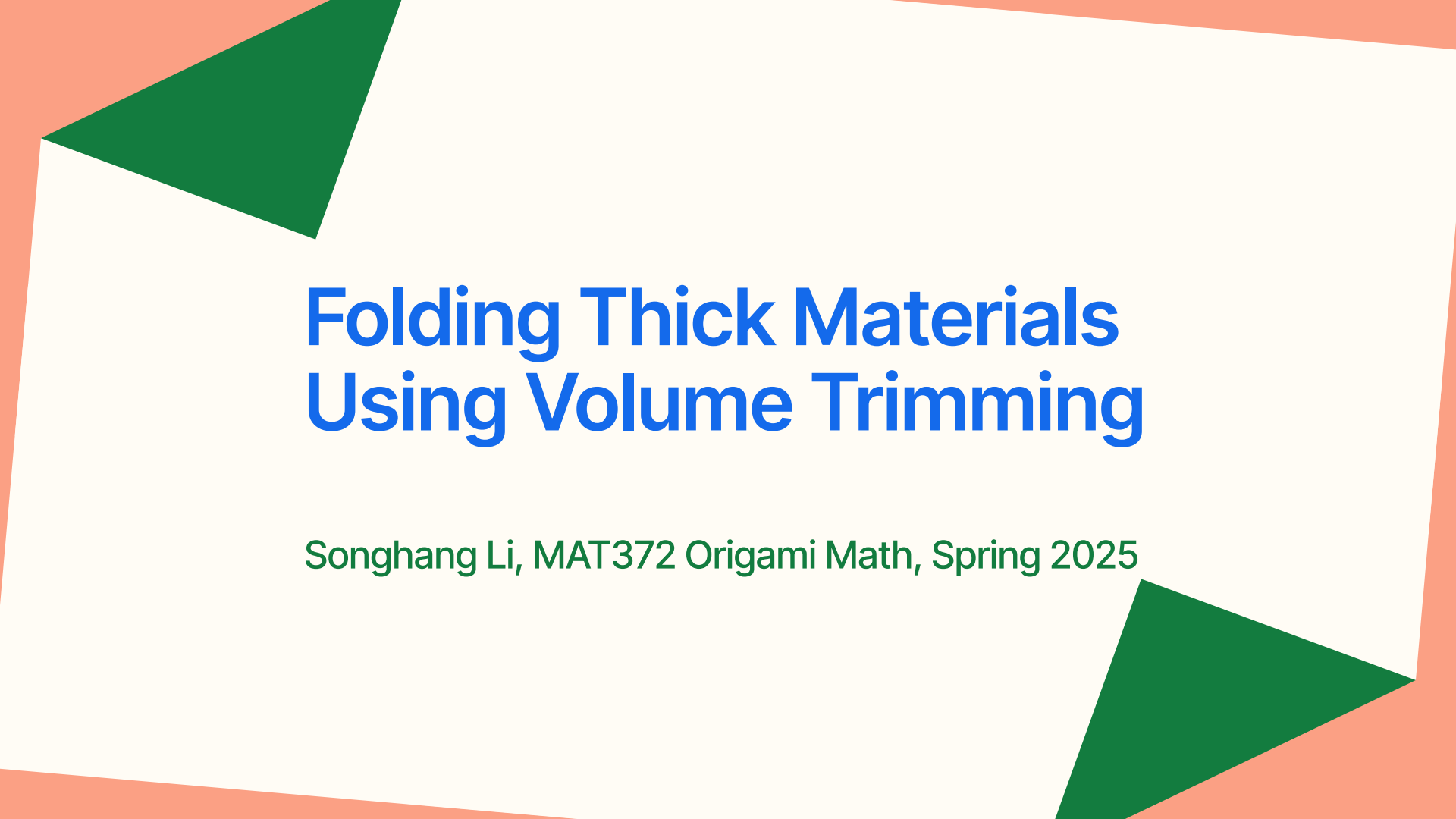
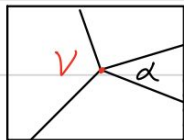




Folding Thick Materials Using Volume Trimming

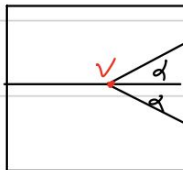
Songhang Li, MAT372 Origami Math, Spring 2025

①



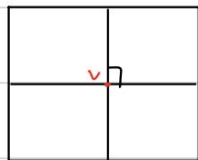
$\exists$ a smallest
angle α

②



$\exists \geq$ smallest
angle $\alpha \quad (\Rightarrow)$ they
are next to each other

③



all 4 angles $\Rightarrow$

Introduction

In class, we've folded degree-4 flat foldable vertices using paper.

But how to fold degree-4 vertex using **thick** material?

Fabrication examples of Acrylic and wood, using laser cutting.

Existing Thick Folding Technique

Hinge Shift

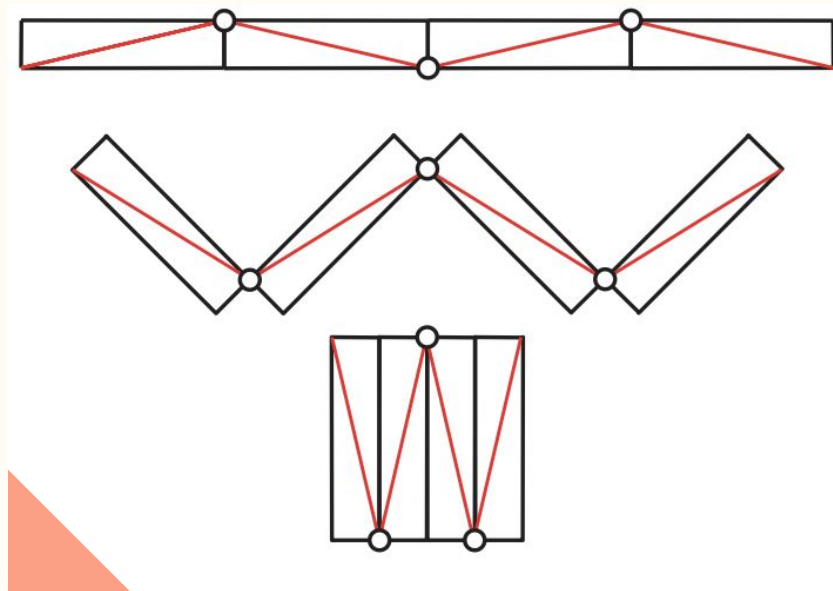
Strained Joint

Offset Panel

Offset Crease

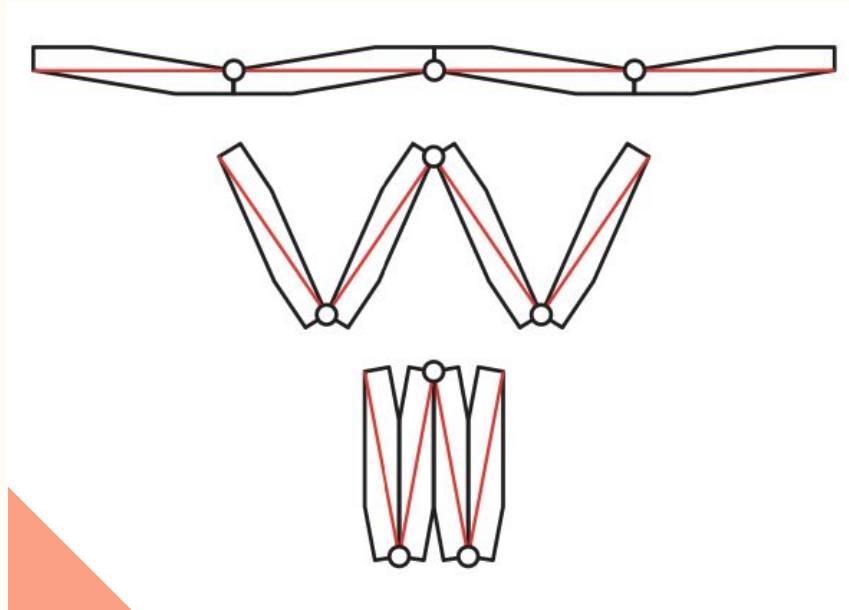
Rolling Contacts

Volume Trimming



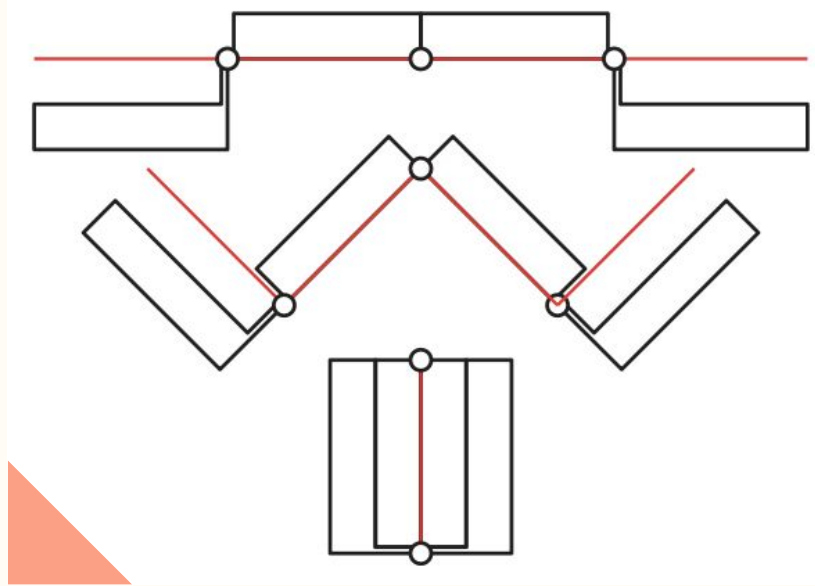
Hinge Shift

Shifts hinges out of plane.
Harder for 2D patterns.



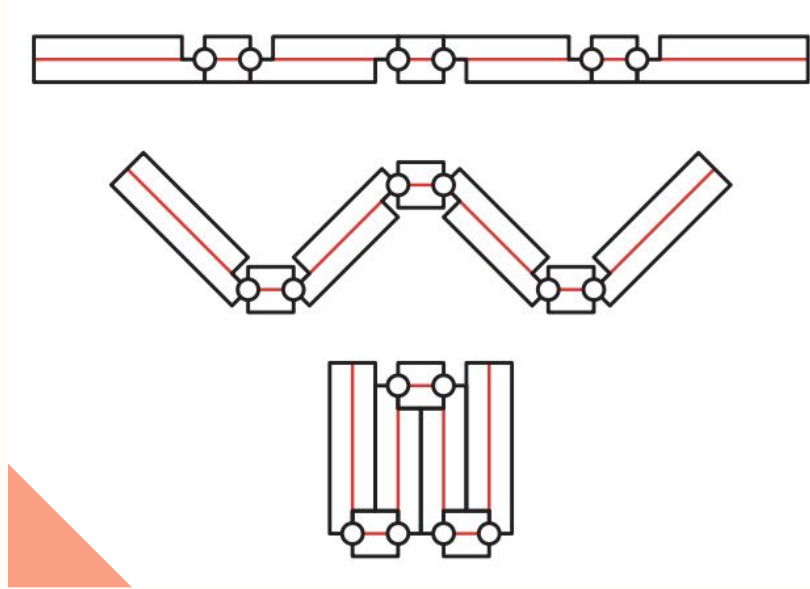
Volume Trimming

Trims the edges of a thickened panel



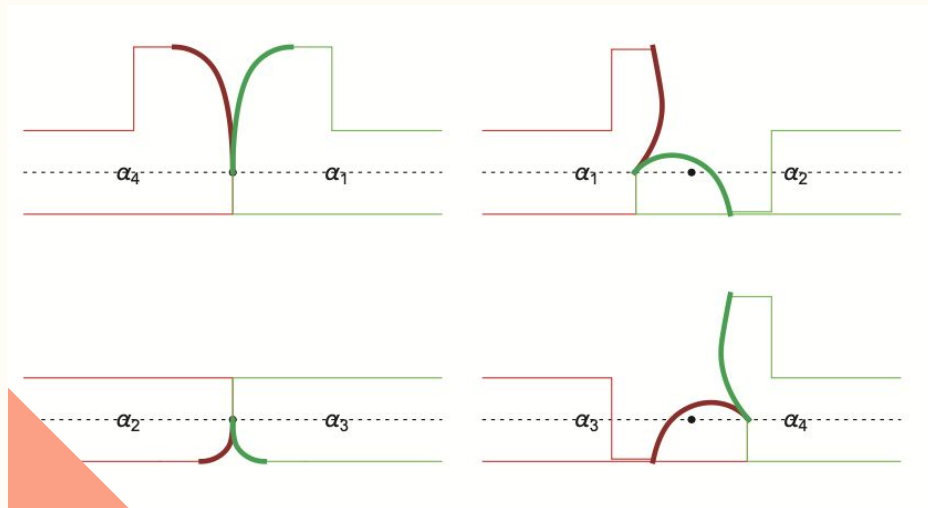
Offset Panel

Shifts panels away from the hinge plane



Offset Crease

Splits each crease into two; allows full motion but adds degrees of freedom. Can be combined with the paper's method

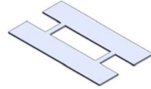


Rolling Contacts

Constructs using elements that roll against each other

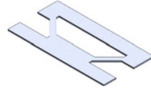
Blending-orthogonal

0



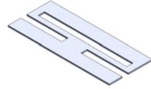
Mixed tension resistant

0–90



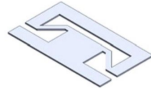
Torsion-parallel

90



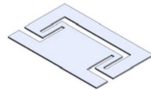
Mixed compression resistant

90–180



Inverted bending-orthogonal

180

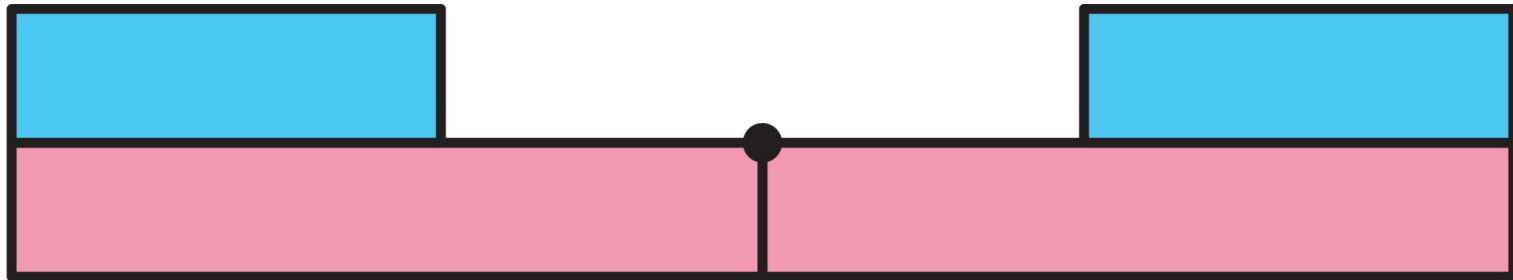


Strained Joint

Selectively removes material to induce hinges; distributes bending over an area

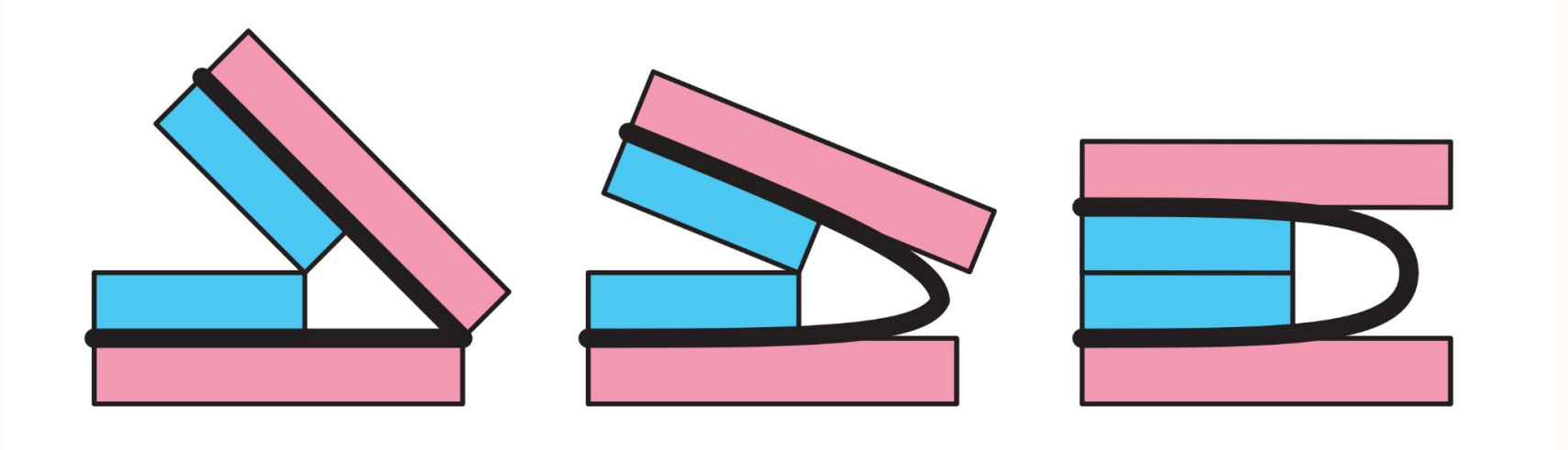
What is Volume Trimming?

Volume Trimming is a technique for folding thick material rigidly by removing interfering volume near the hinge lines.

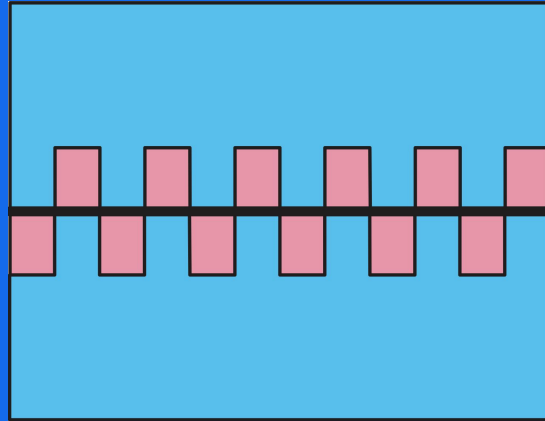


Jason S. Ku

Limitation in practice: Delamination

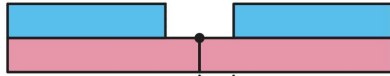


Axially Varying Volume Trimming And Teeth Alternation

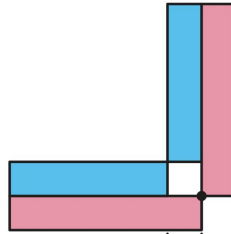


Axially Varying Volume Trimming (Small Angles: $0^\circ - 90^\circ$)

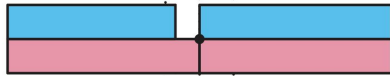
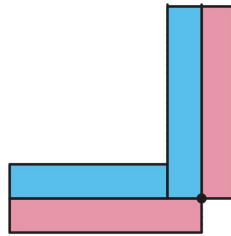
Side



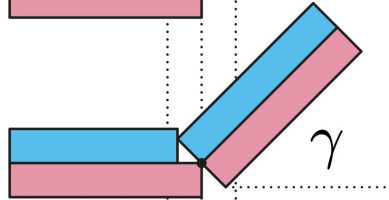
A



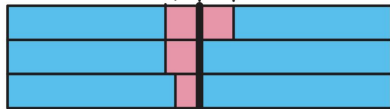
B



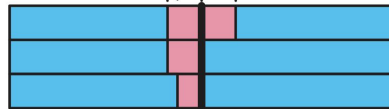
C



Top



A



B



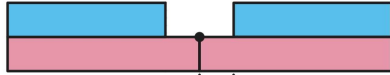
C



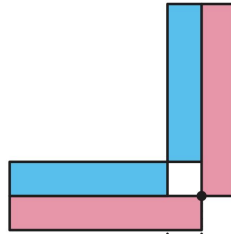
γ

Axially Varying Volume Trimming (Small Angles: $0^\circ - 90^\circ$)

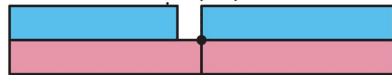
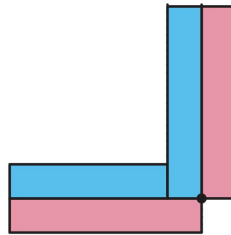
Side



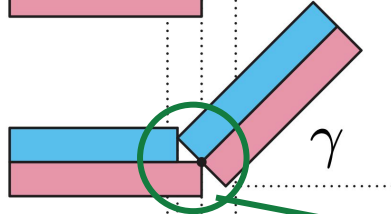
A



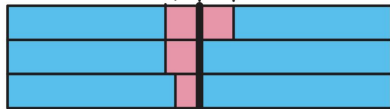
B



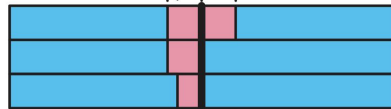
C



Top



A



B



C

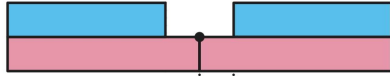


Thickness: t Fold angle: γ

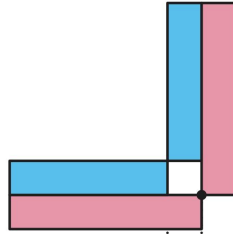
Trimming at distance $t \cot \gamma$ from the fold line

Axially Varying Volume Trimming (with Teeth)

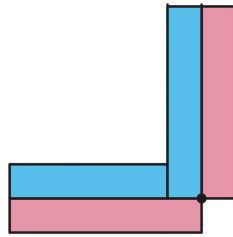
Side



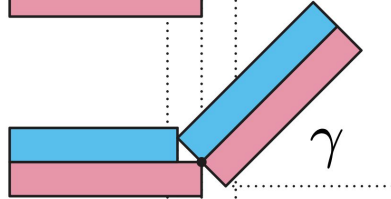
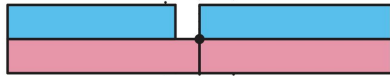
A



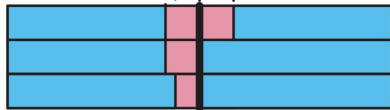
B



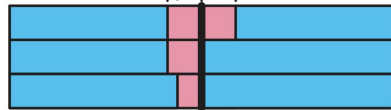
C



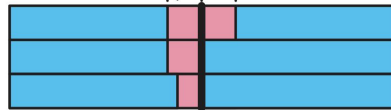
Top



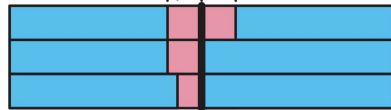
A



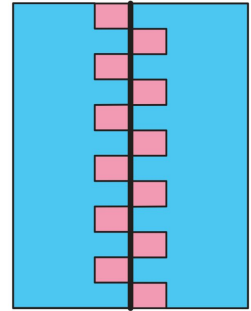
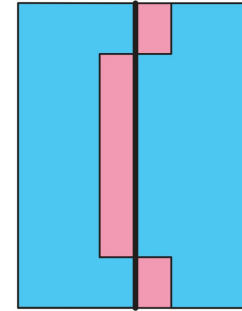
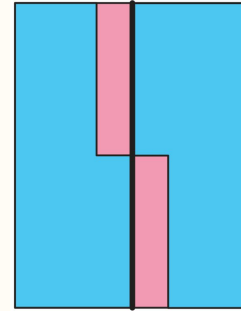
B



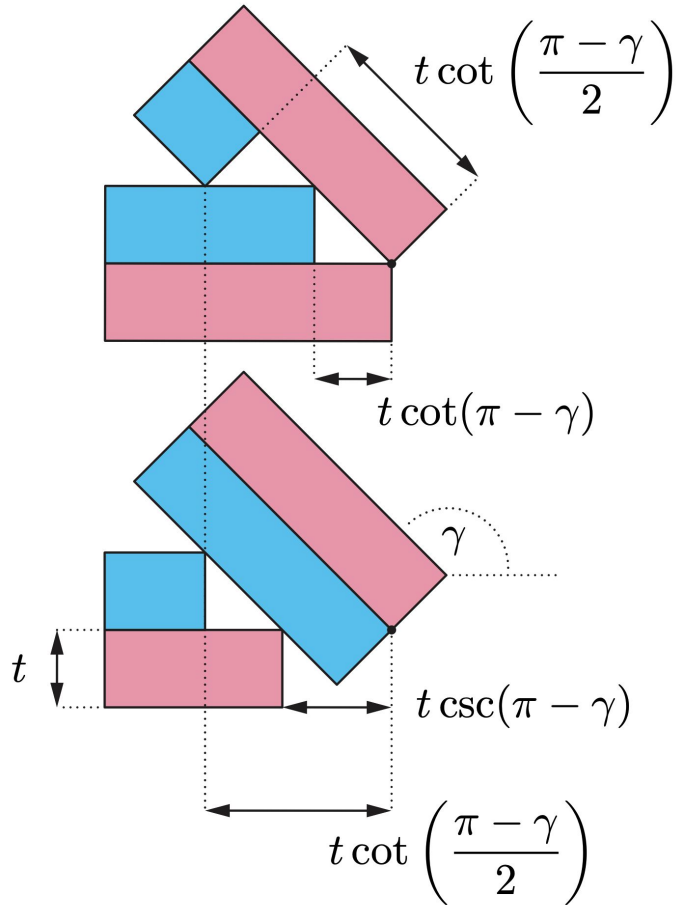
C

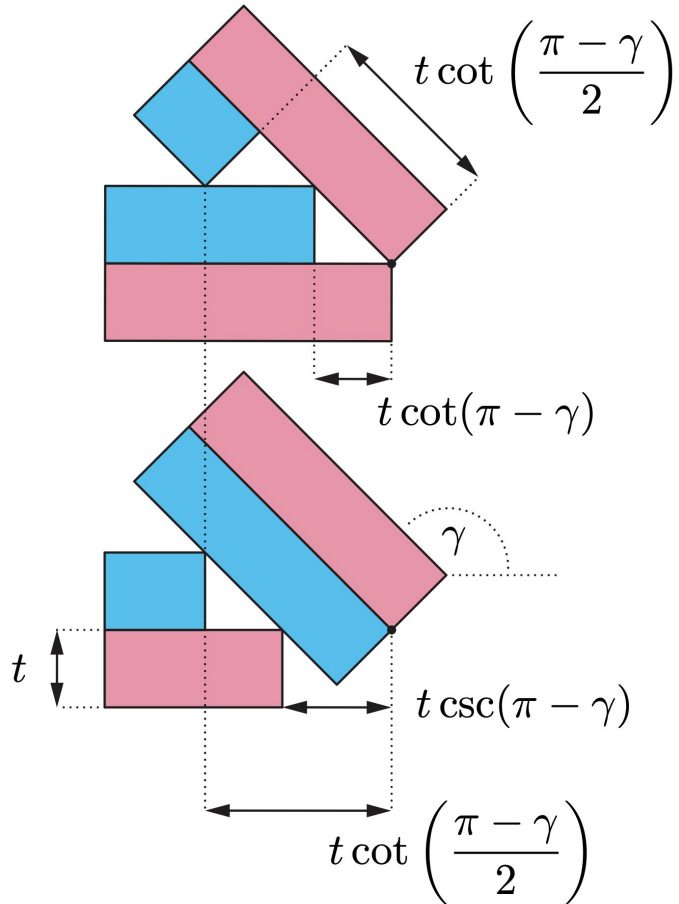


Top



Axially Varying Volume Trimming (Large Angles: 90° - 180°)



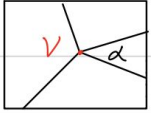


Axially Varying Volume Trimming (Large Angles: $90^\circ - 180^\circ$)

Thickness: t

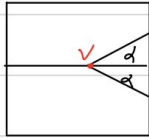
Fold angle: γ

①



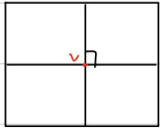
$\exists$ a smallest
angle α

②

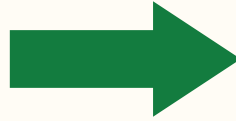


$\exists \geq$ smallest
angle $\alpha \quad (\Rightarrow)$ they
are next to each other

③



all 4 angles $\Rightarrow$

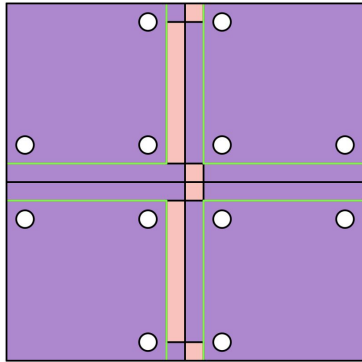
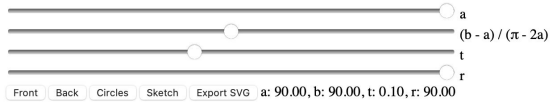


How to fabricate

Design

<http://jasonku.mit.edu/four-crease-hinge-layout/>

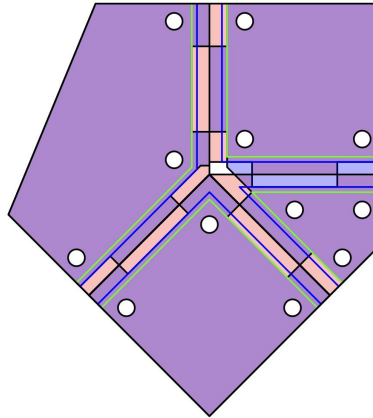
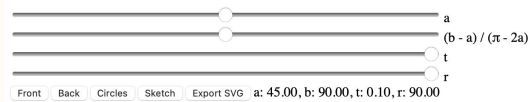
Four Crease Hinge Layout



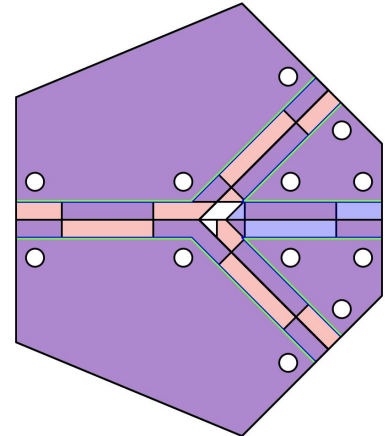
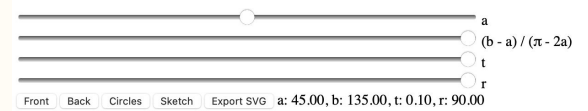
a , the smallest sector angle,

b , the sector angle adjacent to a

r largest fold angle (here restricted to 90)



t , the material thickness



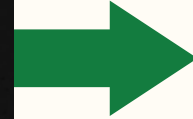
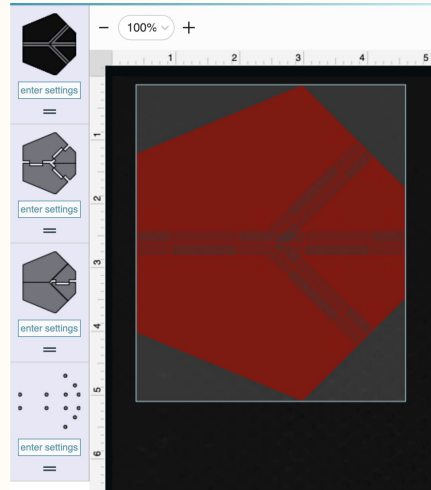
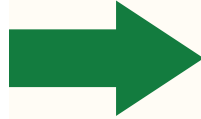
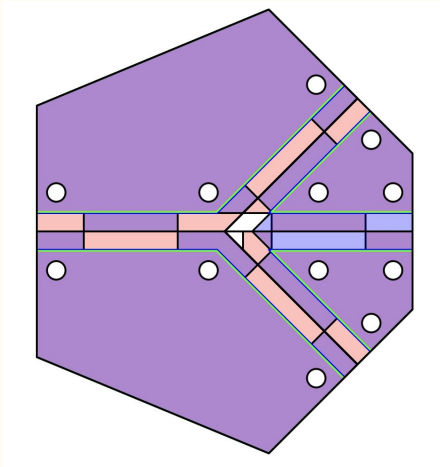
How to fabricate

Design

Laser cutting

Assemble

Front, Back, and Middle





Samples

Reference

- ❑ Ku, J. S. (2017). Folding thick materials via axially varying trimming. ASME IDETC/CIE 2017, Paper DETC2017-67577.
- ❑ Delimont, I. L., Magleby, S. P., & Howell, L. L. (2015). Dual-segment compliant joints as surrogate folds. Journal of Mechanical Design, 137(9).
- ❑ Lang, R. J., Nelson, T., Magleby, S., & Howell, L. (2016). Thick origami using offset rolling contacts. ASME IDETC/CIE 2016, Paper DETC2016-59747.



Thank you